# DIGITAL IMAGE PROCESSING

Comprehensive Question Bank with Solutions

## 2-Mark Questions

### 1. What is a Digital Image?

A digital image is a two-dimensional representation of a scene stored in a computer as a collection of pixels arranged in rows and columns.

### 2. What is Digital Image Processing?

Digital Image Processing (DIP) is the process of using a computer to analyze, manipulate, enhance, and extract information from digital images.

### 3. List any four steps of Digital Image Processing.

1. Image Acquisition
2. Image Enhancement
3. Image Segmentation
4. Image Representation and Description

### 4. What is Morphological Processing and Segmentation?

**Morphological Processing:** A technique used to process image shapes using operations such as erosion and dilation.

**Segmentation:** The process of dividing an image into meaningful regions or objects.

### 5. What is Image Enhancement and Image Restoration?

**Image Enhancement:** Improves the visual appearance of an image.

**Image Restoration:** Removes degradation such as noise and blur to recover the original image.

### 6. Define grayscale image. How are gray-scale images represented?

A grayscale image contains only shades of gray ranging from black to white. It is represented by a matrix where each pixel has an intensity value from 0 (black) to 255 (white).

### 7. What is an indexed image? Write its components.

An indexed image stores pixel values as indices that refer to colors in a color map.

**Components:**

1. Image Matrix (indices)
2. Colormap (color lookup table)

## 8. What is image matrix representation?

An image is represented as a matrix of pixel values arranged in rows and columns, where each element corresponds to the intensity or color of a pixel.

## 9. What is pixel intensity?

Pixel intensity is the brightness value of a pixel in an image. It indicates how dark or bright the pixel appears.

## 10. Define Scotopic, Photopic and Mesopic vision.

**Scotopic Vision:** Vision under very low light conditions (night vision).

**Photopic Vision:** Vision under bright light conditions (daylight vision).

**Mesopic Vision:** Vision under intermediate lighting conditions such as dawn or dusk.

## 11. What is brightness adaptation?

Brightness adaptation is the ability of the human eye to adjust to different levels of illumination.

## 12. Define intensity discrimination.

Intensity discrimination is the ability of the human eye to distinguish between different brightness levels.

## 13. Define Sampling and Quantization.

**Sampling:** Converting a continuous image into discrete pixel locations.

**Quantization:** Assigning discrete intensity values to sampled pixels.

## 14. What is RGB image representation?

RGB representation uses three color components: Red (R), Green (G), and Blue (B). Different combinations of these components produce various colors.

## 15. What is HSV color model?

HSV stands for Hue, Saturation, and Value. It represents colors based on color type, purity, and brightness.

## 16. What is YCbCr model?

YCbCr is a color model where:

$$Y = \text{Luminance (brightness)}$$

$$Cb = \text{Blue chrominance component}$$

$$Cr = \text{Red chrominance component}$$

## 17. What is image cropping?

Image cropping is the process of removing unwanted portions of an image to focus on a specific area.

### 18. What is image resizing?

Image resizing changes the dimensions (width and height) of an image to make it larger or smaller.

### 19. What is image complementing?

Image complementing is the process of reversing pixel intensity values. Bright pixels become dark and dark pixels become bright.

### 20. Define 4-neighbors and 8-neighbors of a pixel.

For a pixel:

**4-neighbors:** Includes the pixels located directly above, below, left, and right of the pixel.

**8-neighbors:** Includes the 4-neighbors plus diagonal neighbors.

### 21. What is adjacency and connectivity?

**Adjacency:** Two pixels are adjacent if they are neighbors and have similar properties.

**Connectivity:** A set of connected pixels forms a connected region in an image.

### 22. List any four applications of Digital Image Processing.

1. Medical Image Analysis
2. Satellite and Remote Sensing Images
3. Face Recognition Systems
4. Industrial Quality Inspection

## Unit II

### 1. What is Intensity Transformation?

Intensity transformation is the process of modifying the intensity values of pixels in an image to improve its appearance or highlight specific features.

### 2. What is Gray Level Transformation? Write its function.

Gray level transformation modifies the gray levels of an image to enhance its quality.

**Function:**

$$s = T(r)$$

Where:

$r$ = input gray level

$s$ = output gray level

$T$ = transformation function

### 3. Define Image Negative.

An image negative is obtained by reversing the gray levels of an image, making bright areas dark and dark areas bright.

**Formula:**

$$s = L - 1 - r$$

Where $L$ is the maximum gray level.

### 4. What is Log Transformation?

Log transformation expands dark pixel values and compresses bright pixel values to improve image details.

**Formula:**

$$s = c \cdot \log(1 + r)$$

where $c$ is a constant.

### 5. Define Power-Law Transformation.

Power-law (Gamma) transformation adjusts image brightness using a power function.

**Formula:**

$$s = c \cdot r^{\gamma}$$

where $c$ and $\gamma$ are constants.

### 6. What is Contrast Stretching?

Contrast stretching is a technique used to increase the range of intensity values in an image, thereby improving its contrast.

### 7. Define Thresholding.

Thresholding is a segmentation technique that converts a grayscale image into a binary image based on a threshold value.

### 8. What is Intensity Level Slicing?

Intensity level slicing highlights a specific range of gray levels while suppressing other intensity values.

### 9. What is Bit-Plane Slicing?

Bit-plane slicing separates an image into a set of binary images, each representing one bit position of the pixel values.

### 10. What is Histogram Stretching?

Histogram stretching expands the range of intensity values in an image to improve contrast and visibility.

### 11. Define Histogram Processing.

Histogram processing is the technique of analyzing and modifying the histogram of an image to enhance its quality.

### 12. What is Histogram Equalization?

Histogram equalization is a method that redistributes pixel intensities to obtain a more uniform histogram and improve image contrast.

### 13. What is Spatial Domain Processing?

Spatial domain processing involves directly manipulating pixel values in an image to perform enhancement operations.

### 14. Define Spatial Filtering.

Spatial filtering is the process of modifying pixel values using a small neighborhood called a mask or kernel.

### 15. What is a Filter Mask?

A filter mask is a small matrix of coefficients used to perform filtering operations on an image.

### 16. Define Correlation.

Correlation is a filtering operation in which a mask is moved across an image and the corresponding pixel values are multiplied and summed without flipping the mask.

### 17. Define Convolution.

Convolution is a filtering operation similar to correlation, but the mask is rotated by 180° before multiplication and summation.

### 18. What is Image Smoothing?

Image smoothing is the process of reducing noise and small variations in an image to produce a smoother appearance.

### 19. What is Averaging Filter?

An averaging filter replaces each pixel value with the average of its neighboring pixel values, thereby reducing noise.

### 20. What is Low-Pass Filter?

A low-pass filter allows low-frequency components to pass and suppresses high-frequency components, resulting in image smoothing and noise reduction.

## Unit III

### 1. Write the structure of RGB color model.

The RGB color model represents colors using three primary components: R (Red), G (Green), and B (Blue).

A color pixel is represented as: $(R, G, B)$ where each component typically ranges from 0 to 255.

### 2. What is Color Models? Write the purpose of Color Models.

**Color Model:** A mathematical model used to represent colors in a standard format.

**Purpose:**

- To represent colors digitally.
- To facilitate image processing and display.
- To convert colors between different devices.

### 3. What is Pseudo Color Image Processing?

Pseudo color image processing assigns colors to grayscale images based on pixel intensity values to improve visualization and interpretation.

### 4. Write the formulae to convert RGB color space into HSV color space.

Let:

$$MAX = max(R, G, B)$$
$$MIN = min(R, G, B)$$
$$V = MAX$$
$$S = (MAX - MIN) / MAX$$

**Hue (H):**

$$H = 60 \times (G - B) / (MAX - MIN) \quad \text{if } MAX = R$$
$$H = 60 \times [2 + (B - R) / (MAX - MIN)] \quad \text{if } MAX = G$$
$$H = 60 \times [4 + (R - G) / (MAX - MIN)] \quad \text{if } MAX = B$$

### 5. Write the formulae to convert RGB color space into NTSC color space and NTSC to RGB.

**RGB to NTSC:**

$$Y = 0.299R + 0.587G + 0.114B$$

$$I = 0.596R - 0.275G - 0.321B$$

$$Q = 0.212R - 0.523G + 0.311B$$

**NTSC to RGB:**

$$R = Y + 0.956I + 0.621Q$$

$$G = Y - 0.272I - 0.647Q$$

$$B = Y - 1.106I + 1.703Q$$

### 6. Write the formulae to convert RGB color space into YCbCr color space and YCbCr to RGB.

**RGB to YCbCr:**

$$Y = 0.299R + 0.587G + 0.114B$$

$$Cb = -0.1687R - 0.3313G + 0.5B + 128$$

$$Cr = 0.5R - 0.4187G - 0.0813B + 128$$

**YCbCr to RGB:**

$$R = Y + 1.402(Cr - 128)$$

$$G = Y - 0.344(Cb - 128) - 0.714(Cr - 128)$$

$$B = Y + 1.772(Cb - 128)$$

### 7. What is YCbCr Color Space?

YCbCr is a color model that separates image information into Y (Luminance/brightness), Cb (Blue chrominance), and Cr (Red chrominance). It is widely used in image and video compression.

### 8. What is HSV?

HSV stands for H (Hue/color), S (Saturation/purity), and V (Value/brightness). It is a color model that represents colors similar to human perception.

### 9. What is CMY?

CMY stands for C (Cyan), M (Magenta), and Y (Yellow). It is a subtractive color model mainly used in color printing.

### 10. What is Image Segmentation?

Image segmentation is the process of dividing an image into meaningful regions or objects for easier analysis.

### 11. Write the mask structure of Sobel Edge Detector.

**$G_x$:**

$$\begin{vmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{vmatrix}$$

**$G_y$:**

$$\begin{vmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{vmatrix}$$

### 12. Write the mask structure of Prewitt Edge Detector.

**$G_x$:**

$$\begin{vmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{vmatrix}$$

**$G_y$:**

$$\begin{vmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{vmatrix}$$

### 13. Write the mask structure of Canny Edge Detector.

Canny edge detector does not use a single mask. It uses a Gaussian filter followed by gradient operators. A commonly used Gaussian mask is:

$$\begin{vmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{vmatrix}$$

### 14. Write the mask structure of Roberts Edge Detector.

**$G_x$:**

$$\begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}$$

**$G_y$:**

$$\begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix}$$

### 15. What is Gaussian Noise?

Gaussian noise is random noise whose intensity values follow a normal (Gaussian) distribution.

### 16. What is Exponential Noise?

Exponential noise is noise whose probability density function follows an exponential distribution.

### 17. What is Uniform Noise?

Uniform noise is noise in which all intensity values within a specified range occur with equal probability.

### 18. What is Rayleigh Noise?

Rayleigh noise is noise whose intensity values follow a Rayleigh probability distribution and is commonly found in radar images.

### 19. What is Salt and Pepper Noise?

Salt and Pepper noise is impulse noise that appears as random black and white pixels in an image.

### 20. List any 4 Noise Models.

1. Gaussian Noise
2. Uniform Noise
3. Rayleigh Noise
4. Salt and Pepper Noise

## Detailed 5/6 Mark Questions

### 1. Explain the Steps in Digital Image Processing with Diagram. (6 Marks)

Digital Image Processing (DIP) is the process of manipulating digital images using a computer. It improves image quality and extracts useful information from images.

**Steps in Digital Image Processing:**

1. **Image Acquisition:** Image acquisition is the first step in digital image processing. In this stage, an image is captured using a camera, scanner, or sensor. The captured image is converted into digital form for further processing.
2. **Image Enhancement:** Image enhancement improves the visual appearance of an image. Techniques such as contrast adjustment, sharpening, and brightness correction are used. This makes important image details more visible.
3. **Image Restoration:** Image restoration removes degradations such as noise and blur from an image. Mathematical models are used to recover the original image. It helps improve image quality without losing important information.
4. **Color Image Processing:** This stage deals with processing color information in images. Color models such as RGB, HSV, and YCbCr are commonly used. It helps in image analysis and object recognition.
5. **Image Segmentation:** Segmentation divides an image into meaningful regions or objects. It separates the object of interest from the background. This simplifies image analysis and interpretation.
6. **Representation and Description:** After segmentation, objects are represented in a suitable format. Features such as shape, size, and texture are extracted. These features help in identifying and classifying objects.
7. **Recognition and Interpretation:** Recognition identifies objects present in the image using extracted features. Interpretation assigns meaning to those objects. This is the final stage where useful decisions are made.

**Diagram:**

```
Image Acquisition → Image Enhancement → Image Restoration → Color Image Processing → Image
```

## 2. Explain Digital Image and RGB Image Representation. (6 Marks)

A digital image is an image stored in digital form and processed using a computer. RGB representation is the most common method used to represent color images.

**Digital Image:**

*Definition:* A digital image is a two-dimensional function $f(x, y)$, where $x$ and $y$ represent spatial coordinates. The value $f(x, y)$ represents the intensity or brightness of a pixel. The image is stored as a matrix of pixels.

*Pixel Representation:* A pixel is the smallest element of a digital image. Each pixel stores intensity information that contributes to the overall image. Millions of pixels together form a complete image.

*Image Matrix:* A digital image can be represented as a matrix of intensity values, for example:

$$\begin{vmatrix} 20 & 40 & 60 \\ 80 & 100 & 120 \\ 140 & 160 & 180 \end{vmatrix}$$

**RGB Image Representation:**

- **Red Component (R):** Represents the intensity of red color in a pixel. Its value ranges from 0 to 255. Higher values indicate a stronger red color contribution.
- **Green Component (G):** Represents the intensity of green color. Different shades of green can be produced by changing its value. It also ranges from 0 to 255.
- **Blue Component (B):** Represents the intensity of blue color in the image. Varying its value produces different shades of blue. Its range is also 0 to 255.

*RGB Pixel:* Each pixel is represented as a triplet: $(R, G, B)$. For example, $(255, 0, 0)$ represents pure red color.

## 3. Explain Grayscale Image and Indexed Image. (6 Marks)

Images can be represented in different formats depending on the application. Grayscale and indexed images are two commonly used image representations in digital image processing.

**Grayscale Image:**

*Definition:* A grayscale image contains only shades of gray ranging from black to white. Each pixel stores a single intensity value. No color information is present in the image.

*Pixel Values:* In an 8-bit grayscale image, pixel values range from 0 to 255. A value of 0 represents black, while 255 represents white. Values between them represent different gray shades.

*Advantages:* Grayscale images require less storage space than color images. They simplify image processing operations and reduce computational complexity. They are widely used in medical imaging and machine vision.

**Indexed Image:**

*Definition:* An indexed image stores pixel values as indices into a color map. Instead of storing actual colors, it stores references to colors. This reduces memory usage.

*Components:* 1. Image Matrix, 2. Color Map (Lookup Table). The image matrix stores index values, while the color map stores actual color information.

*Working:* When an indexed image is displayed, the system uses the stored index value to find the corresponding color in the color map. This method efficiently represents images with limited colors.

*Advantages:* Indexed images save memory and storage space. They are commonly used in GIF images and graphical applications when only a limited number of colors are required.

### 4. Explain Brightness Adaptation and Intensity Discrimination with Diagram. (6 Marks)

The human eye can perceive images under different lighting conditions. Brightness adaptation and intensity discrimination are important properties of the human visual system.

**Brightness Adaptation:**

*Definition:* Brightness adaptation is the ability of the human eye to adjust to different illumination levels. It allows clear vision under both bright and dark conditions. This adaptation occurs automatically.

*Working:* When a person moves from a dark room into sunlight, the eye gradually adjusts to the increased brightness. Similarly, it adapts when moving from a bright area to a dark area. This process helps maintain comfortable vision.

*Importance:* Brightness adaptation enables humans to see over a wide range of light intensities. Without this capability, visibility would be poor under changing lighting conditions.

**Intensity Discrimination:**

*Definition:* Intensity discrimination is the ability of the eye to distinguish between different brightness levels. It helps identify slight changes in image intensity, which is important for perceiving details.

*Working:* The eye can detect differences in brightness only when the intensity difference exceeds a certain threshold. The threshold varies with background illumination. Better discrimination improves image perception.

*Importance:* Intensity discrimination helps in identifying objects and details within an image. It is useful in image enhancement and contrast adjustment techniques.

**Diagram Concept:**

```
Brightness (Perceived)
  ^        /
  |       /
  |      /
  |     /
  +------------------> Illumination Level
```

## 5. Explain Image Representation as a 2D Function. (6 Marks)

A digital image is mathematically represented as a two-dimensional function. This representation helps computers store and process images efficiently.

**Definition:** A digital image is represented as $f(x, y)$, where $x$ is the horizontal coordinate, $y$ is the vertical coordinate, and $f$ is the intensity value at that location.

**Spatial Coordinates:**

- **Horizontal Coordinate (x):** Specifies the horizontal position of a pixel along the width of the image. Every pixel has a unique x-coordinate.
- **Vertical Coordinate (y):** Specifies the vertical position of a pixel along the height of the image. Together with x, it identifies a unique pixel.

**Intensity Values (Brightness Information):** The value $f(x,y)$ represents the brightness of a pixel. Higher values indicate brighter pixels, while lower values indicate darker pixels. In 8-bit grayscale images, values usually range from 0 to 255.

**Matrix Representation:** Mathematically, it is mapped into an grid array:

$$\begin{vmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{vmatrix}$$

**Advantages:**

- *Easy Processing:* The 2D representation allows mathematical operations to be applied easily for operations like image enhancement, filtering, and segmentation.
- *Efficient Storage:* Images can be stored directly as matrices in computer memory, making retrieval and algorithmic manipulation highly efficient.

## 6. Explain Basic Image Processing Operations. (6 Marks)

Basic image processing operations are techniques used to modify, analyze, and improve digital images to extract useful information and enhance image quality.

1. **Image Acquisition:** Capturing an image using a camera, scanner, or sensor and converting it into digital form for computer processing. It is the fundamental first step.
2. **Image Enhancement:** Improving the visual quality of an image using contrast stretching, sharpening, and noise reduction to make specific features more visible.
3. **Image Restoration:** Removing degradations such as blur and noise from an image using mathematical models to recover the original clear state.
4. **Image Compression:** Reducing the storage space or transmission bandwidth required for an image by removing redundant data while maintaining visual quality.
5. **Image Segmentation:** Dividing an image into meaningful regions or distinct objects, separating areas of interest from the background.
6. **Image Recognition:** Identifying and classifying objects present in an image based on extracted features such as shape, color, and texture.

## 7. Explain Image Cropping and Image Resizing. (6 Marks)

Image cropping and image resizing are common image editing operations used to modify dimensions and improve image composition based on application needs.

**Image Cropping:**

*Definition:* The process of removing unwanted outer portions of an image so that only the critical region of interest is retained, focusing attention on core content.

*Working:* A rectangular area is selected from the original image; the outer area is discarded, preserving only the useful chosen section.

*Advantages:* Removes clutter, improves composition, and minimizes overall image data and file size.

**Image Resizing:**

*Definition:* Modifying the geometric dimensions (width and height) of an image, scaling it up (enlarging) or down (reducing) without altering its actual content.

*Working:* Performed using pixel interpolation techniques like nearest-neighbor, bilinear, or bicubic interpolation to compute new pixel values seamlessly.

*Advantages:* Adapts images to fit various display sizes, reduces storage footprints, and accelerates internet transmission speeds.

## 8. Explain Image Complementing with Formula. (6 Marks)

Image complementing is a gray-level transformation technique used to produce the negative of an image. It reverses intensity values and clarifies hidden details in dark regions.

**Definition:** Converts bright pixels into dark ones and dark pixels into bright ones. The output is called the image negative.

**Formula:**

$$s = L - 1 - r$$

Where $r$ is the input pixel intensity, $s$ is the output pixel intensity, and $L$ is the maximum gray level. For a standard 8-bit image, this simplifies to:

$$s = 255 - r$$

**Working Example:**

- Original Value: 0 → Complement Value: 255
- Original Value: 50 → Complement Value: 205
- Original Value: 100 → Complement Value: 155
- Original Value: 255 → Complement Value: 0

**Advantages & Applications:** Enhances visibility in dark areas; highly critical for inspecting X-ray imagery, medical diagnostics, microscopic imaging, and analyzing structural boundaries.

## 9. Explain HSV Color Model. (6 Marks)

The HSV model represents color space based on human color vision components rather than hardware primary color weights.

- **Hue (H):** Represents the actual pure color perceived (e.g., red, green, yellow). It is measured as an angle ranging from 0° to 360°.
- **Saturation (S):** Indicates the purity or strength of the color. High values provide pure vibrant color, while low values drift toward dull gray.
- **Value (V):** Represents the brightness or luminance level of the color, controlling how bright or dark the light appears.

**Key Properties:** It isolates color (chromaticity) from brightness, making it vastly superior to RGB for color segmentation, tracking algorithms, and intuitive user color picking interfaces.

## 10. Explain RGB to Grayscale Conversion. (6 Marks)

RGB images contain red, green, and blue spectral color channels. Converting them into single-channel grayscale simplifies structural processing tasks.

**Conversion Formula:**

$$Gray = 0.299R + 0.587G + 0.114B$$

**Working Principle:** The formula uses weighted coefficients because the human visual system does not perceive all primary colors with equal brightness. The eye is most sensitive to green light, giving it the highest weight ($0.587$), and least sensitive to blue, giving it the lowest weight ($0.114$).

**Example Calculation:** For a pixel with $R=100, G=150, B=200$:

$Gray = 0.299(100) + 0.587(150) + 0.114(200) = 29.9 + 88.05 + 22.8 = 140.75 \approx 141$

**Advantages:** Drastically decreases computational storage and footprint, speeding up downstream machine vision, edge detection, and facial recognition pipelines.

## 11. Explain YCbCr Color Model. (6 Marks)

YCbCr separates image components into a clear brightness channel and separate color-difference components, which is vital for video broadcast and image standard compressions.

- **Y (Luminance):** Carries the pure intensity/brightness information. Most of the critical image structure and detail reside here.
- **Cb (Blue Chrominance):** Represents the difference between the blue component and luminance ($B - Y$).
- **Cr (Red Chrominance):** Represents the difference between the red component and luminance ($R - Y$).

**RGB to YCbCr Conversion equations:**

$$Y = 0.299R + 0.587G + 0.114B$$

$$Cb = -0.1687R - 0.3313G + 0.5B + 128$$

$$Cr = 0.5R - 0.4187G - 0.0813B + 128$$

**Advantages:** Because human eyes notice spatial variations in brightness far better than variations in color, the color components ($Cb, Cr$) can be safely subsampled to dramatically compress JPEG and MPEG streams without visible quality loss.

### 12. Explain Applications of Digital Image Processing. (6 Marks)

Digital Image Processing is integrated into modern fields to improve visualization and fully automate critical workflows:

- **Medical Imaging:** Enhancing and segmenting data from X-rays, MRI, CT scans, and ultrasound systems to help doctors discover anomalies and diagnose critical diseases early.
- **Remote Sensing & Satellites:** Tracking global weather movements, assessing agricultural health, mapping land surfaces, and mapping terrain damage from natural disasters.
- **Industrial Quality Control:** Automating high-speed manufacturing lines to visually check products for scratches, dimensions, and manufacturing assembly flaws.
- **Security & Surveillance:** Powering biometric facial locks, automated object tracking, and discovering suspicious activities in crowds.
- **Traffic Monitoring:** Automatically reading vehicle license plates (ANPR), measuring velocity, and managing real-time congestion in smart cities.
- **Scientific Exploration:** Restoring blurred telescope frames from outer space or sharpening biological samples under high-magnification electron microscopes.

### 13. Explain X-Ray Imaging and CAT Scan. (6 Marks)

X-ray imaging and CAT scans are key diagnostic modalities that reveal internal anatomy non-invasively.

**X-Ray Imaging:** Relies on passing high-energy X-ray beams through a patient's body onto a detector sheet. High-density structures like bone block more radiation and cast a bright silhouette, while soft tissues let rays pass through easily, producing darker regions. It provides quick structural views ideal for identifying bone fractures or lung infections.

**CAT Scan (Computed Axial Tomography):** A highly advanced enhancement where a machine rotates an X-ray source around a patient to capture an array of narrow slices from numerous angles. A computer executes cross-sectional reconstructions to stitch these multi-angle views into detailed 3D volumes of internal soft organs, tumors, and vascular networks.

### 14. Explain Microwave Imaging (Radar). (6 Marks)

Microwave imaging uses high-frequency radio waves to map and sense physical objects or planetary geography.

**Principle & Working:** A radar antenna radiates microwave pulses toward a destination target and listens for returned echoes. By tracking the exact time-of-flight delay and measuring the backscatter echo intensity, a processing system builds highly detailed terrain or spatial layout images detailing object sizes, speeds, and structural shapes.

**Advantages & Uses:** Microwaves easily penetrate environmental blocks like heavy storm clouds, rain, fog, and total nighttime darkness. This makes radar systems highly reliable for weather forecasting, military aircraft navigation, ocean current tracking, and planetary mapping.

## 15. Explain Neighbors of a Pixel with Diagram. (6 Marks)

Analyzing localized spatial operations requires defining structural neighborhoods around a central pixel $P(x, y)$.

- **4-Neighbors ($N_4$(P)):** The four orthogonal elements sharing a straight border with $P$: directly above, below, left, and right.

$$N_4(P) = \{(x-1, y), (x+1, y), (x, y-1), (x, y+1)\}$$

- **Diagonal Neighbors ($N_D$(P)):** The four corner elements sharing only a point intersection with $P$.

$$N_D(P) = \{(x-1, y-1), (x-1, y+1), (x+1, y-1), (x+1, y+1)\}$$

- **8-Neighbors ($N_8$(P)):** The complete full perimeter combination of both orthogonal and diagonal pixels surrounding the center.

$$N_8(P) = N_4(P) \cup N_D(P)$$

**Visual Layout Matrix:**

$$\begin{vmatrix} (x-1,y-1) & (x-1,y) & (x-1,y+1) \\ (x,y-1) & (x,y) & (x,y+1) \\ (x+1,y-1) & (x+1,y) & (x+1,y+1) \end{vmatrix}$$

## 16. Explain Adjacency and Connectivity. (6 Marks)

Adjacency and connectivity establish formal rules to group pixels into distinct objects or continuous structures.

**Adjacency:** Two pixels are adjacent if they are spatial neighbors and their values satisfy a criteria of property similarity (e.g., they belong to the same grayscale gray level range $V$).

- *4-Adjacency:* Two pixels are 4-adjacent if they belong to each other's orthogonal $N_4(P)$ neighborhood.
- *8-Adjacency:* Two pixels are 8-adjacent if they belong to each other's full perimeter $N_8(P)$ block.
- *m-Adjacency (Mixed):* Used to eliminate ambiguous diagonal paths. Two pixels are m-adjacent if they are 4-neighbors, or if they are diagonal neighbors and their intersecting 4-neighbors do not share property values.

**Connectivity:** Describes a continuous path between distant pixels. If a chain of adjacent pixels can be traced from pixel $A$ to pixel $B$, they are defined as connected. This is essential for segmenting text characters, tracing boundaries, and regional parsing.

## 17. Explain Gamma-Ray Imaging and its Applications. (6 Marks)

Gamma-ray imaging uses high-energy electromagnetic nuclear radiation to view internal components or deep material structures.

**Working Principle:** Gamma rays feature short wavelengths and massive material penetration power. In medical scenarios, radioisotopes are injected into the patient. These isotopes collect in active organs and emit gamma rays outward, which are captured by specialized gamma camera tracking sensors to map metabolic action or locate tumors (such as in PET scans).

**Industrial & Scientific Roles:** Used for non-destructive inspection of dense metal structures, thick oil pipelines, and structural welds to spot hidden stress fractures without dismantling components. It also aids astrophysics research in capturing cosmic data from celestial bodies.

## 18. Explain Image Negative and Log Transformation. (6 Marks)

These spatial transformations modify pixel values to map dark and bright features into more visible ranges.

**Image Negative:** Inverts the gray scale spectrum via $s = L - 1 - r$. This turns dark areas white and bright areas black, which is highly beneficial for highlighting structural details hidden within large black backgrounds (e.g., medical film scans).

**Log Transformation:** Expressed as $s = c \cdot \log(1 + r)$. A log curve expands low-intensity dark regions while compressing high-intensity bright regions. This maps large dynamic ranges of input values (such as a Fourier spectrum with enormous intensity spikes) into a visible display scale.

## 19. Explain Intensity Transformation Functions and Power-Law Transformation. (6 Marks)

Intensity functions modify pixel brightness values individually to improve overall visibility and contrast.

**Power-Law (Gamma) Transformation:** Expressed mathematically by the equation:

$$s = c \cdot r^{\gamma}$$

By shifting the exponent value $\gamma$, the transformation maps a wide scale of curves:

- When $\gamma < 1$: The function expands dark gray levels and compresses bright ones, increasing overall brightness.
- When $\gamma > 1$: The function compresses dark values and expands highlight regions, darkening the overall image.

**Applications:** Crucial for performing gamma correction on CRT, LCD, and broadcast displays to ensure captured light levels match real-world perceptions accurately.

## 20. Explain Contrast Stretching. (6 Marks)

Contrast stretching is a linear point transformation used to expand an underexposed, narrow range of input intensities across the full dynamic capability of a display device.

**Working & Visibility:** Low-contrast images often look muddy and gray because their pixels occupy a narrow range of values. Contrast stretching applies a linear mapping to stretch this narrow range across the full 0–255 scale. This increases the structural difference between dark and bright zones, sharpening object outlines and bringing out fine textures in photographs and satellite frames.

## 21. Explain Histogram Processing and its Types. (6 Marks)

An image histogram graphs the total pixel population distribution counted at each discrete gray intensity level, offering global diagnostic views of contrast and exposure.

**Primary Processing Variants:**

- **Histogram Stretching:** Manually scales the minimum and maximum active values to occupy the full spectrum.
- **Histogram Equalization:** Uses an automated global function to redistribute pixel densities into a flat, uniform distribution, maximizing global contrast.
- **Histogram Specification:** Intervenes to shape the intensity profile into a custom target distribution curve.
- **Local Histogram Processing:** Applies contrast-enhancing operations over small local sliding patches to clarify minute details without over-saturating the entire image.

## 22. Explain Histogram Equalization Process with an Example. (6 Marks)

Histogram equalization linearizes the cumulative distribution function (CDF) of an image to maximize its global contrast.

**Algorithmic Implementation Steps:**

1. Count total pixels ($n_k$) at each gray level ($r_k$).
2. Compute the probability of occurrence: $P(r_k) = n_k / N$.
3. Form the Cumulative Distribution Function (CDF) by accumulating these probabilities.
4. Map to new gray levels using the transformation function: $s_k = round[(L - 1) \times CDF]$.

**Example Demonstration:** If an ultra-compact 3x3 low-contrast matrix transitions from an input profile to an equalized output profile, the values map as follows:

Input Matrix Array:

$$\begin{vmatrix} 50 & 50 & 60 \\ 60 & 70 & 80 \\ 80 & 90 & 90 \end{vmatrix}$$

Output Equalized Matrix Array:

$$\begin{vmatrix} 20 & 20 & 60 \\ 60 & 120 & 180 \\ 180 & 240 & 240 \end{vmatrix}$$

The resulting matrix spreads intensities across a wider range, enhancing local contrast and visibility.

## 23. Explain Correlation with an Example. (6 Marks)

Correlation is a spatial neighborhood operation where a small tracking template or filter mask shifts over an image area to compute similarity scores.

**Mathematical Form:**

$$g(x, y) = \sum \sum w(s, t) \cdot f(x + s, y + t)$$

The mask is shifted across the target location without any rotation. Corresponding values are multiplied together and then summed.

**Numerical Example:** Given a local patch and a basic mask:

Image Section Matrix:

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$$

Filter Mask Matrix:

$$\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix}$$

Calculation:

$$Output = (1\times1) + (2\times0) + (3\times1) + (4\times0) + (5\times1) + (6\times0) + (7\times1) + (8\times0) + (9\times1) = 1 + 3 + 5 + 7 + 9 = 25$$

This is commonly used for template matching and feature detection applications.

## 24. Explain Convolution with an Example. (6 Marks)

Convolution is a fundamental filtering operation in image processing, used for smoothing, sharpening, and feature extraction.

**Mechanism:** Convolution follows the same steps as correlation, but the filter mask must be rotated or flipped by 180° before starting the multiplication and accumulation process.

**Mathematical Form:**

$$g(x, y) = \sum \sum w(s, t) \cdot f(x - s, y - t)$$

**Example Mask Rotation:** Given an initial edge-checking mask:

$$\begin{vmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{vmatrix}$$

After a 180° flip, it becomes:

$$\begin{vmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{vmatrix}$$

This flipped mask is then multiplied by the image intensities to generate the output value, forming the baseline for linear filtering algorithms.

## 25. Explain Image Smoothing Using Averaging Filter. (6 Marks)

Averaging filters smooth images and reduce high-frequency noise by blending neighboring pixel intensities.

**Mask Structure:** A standard 3x3 uniform box filter is defined as:

$$W = 1/9 \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

**Working Example:** For a pixel neighborhood containing the following values:

$$\begin{vmatrix} 20 & 30 & 40 \\ 30 & 40 & 50 \\ 40 & 50 & 60 \end{vmatrix}$$

Calculation:

$Average = (20 + 30 + 40 + 30 + 40 + 50 + 40 + 50 + 60) / 9 = 360 / 9 = 40$

The old central value (40) is replaced with the computed average (40), smoothing out sharp noise spikes across the matrix.

## 26. Explain Image Sharpening. (6 Marks)

Image sharpening highlights edges and fine details while suppressing slow spatial variations, improving image clarity.

**Mechanism & Mask:** Sharpening strengthens high-frequency components by emphasizing regions with sudden intensity changes. This is achieved using high-pass kernels, such as the Laplacian mask:

$$\begin{vmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{vmatrix}$$

**Techniques:** Common approaches include Laplacian sharpening, high-pass spatial filtering, and unsharp masking. These methods make boundaries much clearer, which is essential for medical analysis and satellite feature mapping.

## 27. Explain Low-Pass Median Filter (Order Statistic Filter). (6 Marks)

The median filter is a non-linear spatial sorting filter that is highly effective at removing impulsive noise while preserving sharp boundaries.

**Working Steps:** Rather than computing mathematical averages, the filter collects all pixel values within a neighborhood window, sorts them in ascending order, and assigns the exact median (middle) value to the central pixel.

**Numerical Example:** Given a neighborhood containing an extreme noise spike (200):

$$\begin{vmatrix} 10 & 20 & 30 \\ 40 & 200 & 50 \\ 60 & 70 & 80 \end{vmatrix}$$

Sorted Order List: ***10, 20, 30, 40, 50, 60, 70, 80, 200***

The median value is ***50***. The noise spike (200) is replaced with ***50***, removing the salt-and-pepper artifact without blurring the surrounding edges.

## 28. Explain High-Pass Filtering. (6 Marks)

High-pass filtering attenuates low-frequency components to highlight fine details and edge transitions.

**Working Principle:** The filter computes differences between adjacent pixel values, amplifying large intensity transitions while reducing uniform areas to zero. A common high-pass kernel is:

$$\begin{vmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{vmatrix}$$

This operation sharpens object boundaries, making it highly useful for structural analysis and pattern recognition.

## 29. Explain Enhancement Using Arithmetic/Logic Operations. (6 Marks)

Arithmetic and logic operations process images by combining pixel values pixel-by-pixel.

- **Image Addition:** Combines two images via $g(x,y) = f_1(x,y) + f_2(x,y)$. This is often used to average multiple noisy frames together to reduce random noise.
- **Image Subtraction:** Removes shared background information via $g(x,y) = f_1(x,y) - f_2(x,y)$, which is useful for change detection and medical angiography.
- **Image Multiplication:** Multiplies intensities via $g(x,y) = f_1(x,y) \times f_2(x,y)$, typically used for binary masking and region-of-interest gating.
- **Logical Operations (AND, OR, XOR):** Performs bitwise comparisons. AND acts as an intersection mask; OR combines binary features; XOR highlights mismatches between frames.

## 30. Explain Smoothing (Low-Pass) Spatial Filters. (6 Marks)

Smoothing spatial filters suppress high-frequency noise and rapid intensity transitions, producing a softer visual appearance.

**Mechanisms:** These filters apply neighborhood convolution masks, such as box or Gaussian filters, to compute localized weighted averages. While this process effectively reduces random sensor noise, it also softens sharp details and introduces a slight blurring effect across edges.

## 31. Explain Sharpening (High-Pass) Spatial Filters. (6 Marks)

Sharpening filters enhance high-frequency details, such as fine lines and object boundaries, by amplifying spatial derivatives.

**Implementation:** These filters highlight rapid intensity changes using derivative-based kernels, like the Laplacian operator. This increases the contrast along boundaries, making fine features much clearer and easier for computer vision systems to parse.

## 32. Explain Order Statistic (Non-Linear) Filters. (6 Marks)

Order statistic filters are non-linear operators that filter pixels by sorting neighborhood values rather than computing linear combinations.

- **Median Filter:** Replaces the center pixel with the middle value of the sorted list, effectively removing salt-and-pepper noise while preserving sharp boundaries.
- **Max Filter:** Selects the maximum intensity in the window, which helps remove pepper noise and highlight bright features.
- **Min Filter:** Selects the minimum intensity in the window, which helps remove salt noise and preserve dark regions.
- **Midpoint Filter:** Computes the average of the minimum and maximum values: $Midpoint = (Max + Min) / 2$. This is highly effective at reducing uniform or Gaussian noise distributions.

## 33. Explain RGB and HSV Color Models. (6 Marks)

**RGB Model:** An additive color space based on mixing primary components: Red, Green, and Blue. Each channel typically ranges from 0 to 255. It matches the hardware design of cameras and monitors, but it does not align well with how humans perceive and describe color.

**HSV Model:** Organizes color information into three intuitive channels: Hue (the core color angle), Saturation (the color's purity), and Value (the brightness level). By separating brightness from chromaticity, the HSV model is widely used for color-based object segmentation and tracking tracking tasks.

## 34. Explain NTSC and CMY Color Models. (6 Marks)

**NTSC Model:** Used in television broadcasting, this model formats color into Y (Luminance/brightness), I (In-phase color component), and Q (Quadrature color component). Separating brightness from color improves transmission efficiency and ensures compatibility with legacy black-and-white television systems.

**CMY Model:** A subtractive color model based on Cyan, Magenta, and Yellow. It is primarily used in color printing, where colors are produced by subtracting specific wavelengths of light from a white paper background: $C = 255-R, M = 255-G, Y = 255-B$.

## 35. Explain Pseudo Color Image Processing. (6 Marks)

Pseudo color processing maps grayscale intensities to specific colors using a lookup table, enhancing human visual interpretation without altering the underlying data.

**Application Example:** Low intensities can be mapped to cool colors like blue, while high intensities map to warm colors like red. This technique is widely used in thermal imaging, satellite analysis, and medical scans to help the human eye spot subtle variations that are hard to distinguish in grayscale.

## 36. Detect Edges of Given Image Using Prewitt Edge Detector Algorithm. (6 Marks)

The Prewitt edge detector is a gradient-based operator that identifies boundaries by measuring horizontal and vertical intensity changes.

**Mask Structures:**

Horizontal Operator Matrix ($G_x$):

$$\begin{vmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{vmatrix}$$

Vertical Operator Matrix ($G_y$):

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{vmatrix}$$

**Algorithmic Steps:** Both masks convolve across the frame to compute independent directional gradients ($G_x$ and $G_y$). The total edge strength at each pixel is determined by calculating the gradient magnitude: $G = \sqrt{(G_x^2 + G_y^2)}$. Pixels with high magnitude scores are identified as edge boundaries.

## 37. Detect Edges of Given Image Using Roberts Edge Detector. (6 Marks)

The Roberts edge detector is an early, computationally efficient operator that calculates gradients along diagonal directions.

**Mask Structures:** Uses compact 2x2 spatial templates:

$$G_x = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} \quad G_y = \begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix}$$

**Processing Steps:** Convolves both 2x2 matrices to find diagonal transitions. The total gradient magnitude is computed as: $G = \sqrt{(G_x^2 + G_y^2)}$. While highly sensitive to high-frequency noise, it remains useful for extracting sharp, distinct boundaries.

## 38. Detect Edges of Given Image Using Canny Edge Detector. (6 Marks)

The Canny Edge Detector is a highly reliable, multi-stage pipeline designed to ensure clean, accurate edge detection while minimizing noise artifacts.

**Algorithmic Pipeline Stages:**

1. **Gaussian Smoothing:** Convolves the image with a Gaussian blur filter to remove high-frequency noise and prevent false edge detections.
2. **Gradient Computation:** Applies derivative operators to calculate horizontal and vertical gradients, determining edge strength and direction.
3. **Non-Maximum Suppression:** thins broad responses by suppressing any gradient values that are not local maxima along the edge direction, leaving thin, sharp lines.
4. **Double Thresholding:** Separates candidate pixels into strong edges (above a high threshold) and weak edges (between low and high thresholds).
5. **Hysteresis Edge Tracking:** Preserves weak edges only if they are connected to strong edges, filtering out isolated noise points.

## 39. Explain Sobel Edge Detector. (6 Marks)

The Sobel Edge Detector is a gradient-based operator that incorporates a smoothing effect to reduce noise while detecting boundaries.

**Mask Structures:** Uses 3x3 directional kernels:

Horizontal Edge Checking Matrix ($G_x$):

$$\begin{vmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{vmatrix}$$

Vertical Edge Checking Matrix ($G_y$):

$$\begin{vmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{vmatrix}$$

The central weight of 2 introduces a localized smoothing effect. The final gradient magnitude is computed as: $G = \sqrt{(G_x^2 + G_y^2)}$. This provides robust edge responses, making it a standard choice for image segmentation pipelines.

## 40. Explain Gaussian Noise. (6 Marks)

Gaussian noise is a statistical model where the noise intensity values follow a standard normal (Gaussian) distribution, typically caused by electronic circuit heat and sensor noise.

**Probability Density Function (PDF):**

$$p(z) = [1 / \sqrt{(2\pi\sigma^2)}] \cdot e^{-(z-\mu)^2 / (2\sigma^2)}$$

Where $\mu$ represents the mean intensity and $\sigma$ represents the standard deviation. It adds a fine, grainy texture across every pixel, making it highly valuable for benchmarking and testing image restoration filters.

## 41. Explain Exponential Noise. (6 Marks)

Exponential noise is a random noise model where the probability density function drops off sharply as the intensity value increases.

**Probability Density Function (PDF):**

$$p(z) = a \cdot e^{-az} \quad \text{for } z \geq 0$$

This distribution means low-intensity noise values occur much more frequently than high-intensity ones. It is widely used in simulation studies to evaluate how well noise-removal filters perform under asymmetrical noise profiles.

## 42. Explain Uniform Noise. (6 Marks)

Uniform noise features a flat probability distribution, meaning every intensity value within a defined range has an equal chance of occurring.

**Probability Density Function (PDF):**

$$p(z) = 1 / (b - a) \quad \text{if } a \leq z \leq b$$

$$p(z) = 0 \quad \text{otherwise}$$

This noise introduces random, evenly distributed variations across the entire image. It is commonly used in synthetic modeling, signal processing simulations, and computer graphics testing.

## 43. Explain Rayleigh Noise. (6 Marks)

Rayleigh noise features an asymmetrical distribution that rises to a peak before dropping off gradually. It typically appears in radar, sonar, and ultrasound imaging systems.

**Probability Density Function (PDF):**

$$p(z) = [(z - a) / b] \cdot e^{-(z - a)^2 / (2b)} \quad \text{for } z \geq a$$

This model is valuable for simulating real-world coherent imaging conditions, helping researchers optimize restoration algorithms for specialized sensor systems.

### 44. Explain Salt and Pepper Noise. (6 Marks)

Salt and pepper noise is a form of impulsive noise that appears as isolated, random black and white pixels, often caused by transmission errors or faulty sensor elements.

**Characteristics & Example:** Noisy pixels differ sharply from their neighbors, taking on maximum brightness (255 for salt) or minimum darkness (0 for pepper).

Input Noisy Matrix Example:

$$\begin{vmatrix} 50 & 255 & 50 \\ 0 & 50 & 50 \\ 50 & 50 & 255 \end{vmatrix}$$

Here, the values 255 and 0 represent salt and pepper artifacts, respectively. Non-linear median filters are highly effective at removing this noise while keeping surrounding edges sharp.

## 45. Calculate Histogram Equalization Process for Given Gray Levels. (6 Marks)

**Given Input Data Table:**

| Gray Level ($r_k$) | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| No. of Pixels ($n_k$) | 790 | 1023 | 850 | 656 | 329 | 245 | 122 | 81 |

**Step 1: Compute Total Number of Pixels (N) & Gray Levels (L)**

$N = 790 + 1023 + 850 + 656 + 329 + 245 + 122 + 81 = 4096$

Total grey levels $L = 8$, so the scaling factor $(L - 1) = 7$.

**Step 2 & 3: Calculate Probability $P(r_k)$ and Cumulative Distribution Function (CDF)**

| $r_k$ | $n_k$ | $P(r_k) = n_k / N$ | CDF | $S_k = 7 \times$ CDF | Rounded Output Gray Level |
|---|---|---|---|---|---|
| 0 | 790 | 0.193 | 0.193 | 1.35 | **1** |
| 1 | 1023 | 0.250 | 0.443 | 3.10 | **3** |
| 2 | 850 | 0.208 | 0.651 | 4.56 | **5** |
| 3 | 656 | 0.160 | 0.811 | 5.68 | **6** |
| 4 | 329 | 0.080 | 0.891 | 6.24 | **6** |
| 5 | 245 | 0.060 | 0.951 | 6.66 | **7** |
| 6 | 122 | 0.030 | 0.981 | 6.87 | **7** |
| 7 | 81 | 0.020 | 1.000 | 7.00 | **7** |

**Final Histogram Equalization Mapping Result:**

Original levels **[0, 1, 2, 3, 4, 5, 6, 7]** map directly onto new equalized gray values **[1, 3, 5, 6, 6, 7, 7, 7]**. This redistribution expands the dynamic range, enhancing contrast and making fine image details much clearer.